

Eric Du

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EXPERIENCE

Software Engineer

Apr 2024 – Aug 2026

Symmetry Laser — Davis, CA

Sole software engineer for a 50-person manufacturer across two sites.

- Owned every internal web application for a **50-person** manufacturer across two sites, feeding **36 floor monitors** with live job and station scheduling.
- Cut bundle size from **5MB to 1.3MB** on the low-powered PCs driving **36 floor displays** on a 5-second polling loop, rewriting the internal web platform from React to Svelte.
- Cut p50 query time from **4s to 10ms** on 90% of queries by replicating 36 FileMaker tables into PostgreSQL and moving 12+ services onto the replica.
- Cut quote turnaround from **40-60 minutes to 5-7** on 20+ quotes a week with a RAG pipeline on AWS Bedrock that reads RFQ emails and engineering drawings.
- Containerized 8 services behind self-hosted Gitea CI/CD, holding **99.9% uptime**, ending 5-6 manual restarts a month, and cutting deploys from **20 minutes to 1**.

Full Stack Engineer

Apr 2023 – Apr 2024

UC Davis, Department of Plant Sciences — Davis, CA

- Built FastAPI and AWS Lambda services predicting planting dates from local weather station data, surfaced in a Vue component on Drupal.

Full Stack Engineer

Sep 2022 – Dec 2023

AggieWorks — Davis, CA

Student-run software team building products for the UC Davis community.

- Cut image load times **15x** and built a 23-feature matching algorithm for RoomU, a React Native and Go app with **120+ users**.
- Built messaging, pagination, and image caching for AggieExchange, a Next.js marketplace with **160+ users**.

PROJECTS

Costco Gas Tracker | React, FastAPI, Redis, TimescaleDB, Docker | github.com/EricDooo/costco-pump Aug 2026

- Tracks live fuel prices at every Costco station across **11 countries**, reconciling three undocumented APIs across two Costco platforms into one feed with full price history in a TimescaleDB hypertable.
- Split ingestion into independent Redis/RQ jobs so one slow country never blocks the rest, and self-hosted the Protomaps basemap to run with **no third-party map API key or quota**.

Fish Species Classifier | PyTorch, Django

Sep 2023

- Matched baseline accuracy with **25% of ResNet-50's parameters** using a 6M-parameter CNN, placing **2nd of 36 teams**.
- Served predictions through a Django app reporting per-class accuracy on uploaded images.

TECHNICAL SKILLS

Frontend: React, SvelteKit, Vue.js, Next.js, React Native, Qt/PySide6

Languages: Rust, Python, C#, Go, TypeScript, JavaScript, SQL, Bash, Structured Text (IEC 61131-3)

Backend: Axum, FastAPI, Django, Node.js, REST APIs

Data: PostgreSQL, TimescaleDB, PostGIS, MySQL, InfluxDB, FileMaker, Apache Airflow, Redis

Infrastructure and DevOps: Docker, Gitea, GitHub Actions, CI/CD, AWS (Bedrock, Lambda, S3), Git

EDUCATION

University of California, Davis — Davis, CA

Sep 2020 – Dec 2023

Bachelor of Science in Computer Science | GPA: 3.7/4.0

Relevant coursework: Deep Learning, Computer Vision, Operating Systems, Computer Networks, Databases, System Design